ITA-0518 - Computer Vision

Co2 - AT3 - Class Test

Name: Dhaanush Rajan
Regno: 192421162

1) An image appears too bright.
Explain and justify the most suitable grey level transformation to improve visibility. Clearly describing the concept application process and expected outcome.

2) An dark images lacks details.
Describe and justify an appropirate grey level transformation to enhance visibility, explaining howit improves low-intensity region with proper reasoning

3) An image has poor contrast.
Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning

4) A low-contrast image requires uniform intensity distribution
Describe and justify the method you would apply, explaining the concept working process and its effect on intensity levels

5) An image shows uneven brightness across regions.
Explain and apply an appropriate enhancement technique to correct non-uniform illumination Providing a structured and accurate explaination

MTA-0578 - Computer Vision

CO2-AT3-ClassTest

1) An image appears too bright the most suitable gray level transformation is the Negative (Brightness Reduction using Intensity Inversion) if the goal is to emphasize dark details, or more commonly Power-Law Gama Correction ($\gamma > 1$) is generally

Grey Level Transformation : Gamma (Power-Law) Transformation

Gamma transformation modifies the intensity value of an image according to the equation

$$s = cr^{\gamma}$$

r = Input pixel intensity
s = output pixel intensity
c = constant
γ = gamma value

When $\gamma > 1$, bright pixels become darker, reducing the image

Expected outcome:-

i) Excessive brightness is reduced.
i) Details in overexposed region become more visible
i) Overall image contrast is improved
i) The image appears more natural and easier to interpret

Justification:-

- Reduces brightness without loosing important image information.
- Preserves contrast better than simply subtracting a constant intensity.
- Is widely used for image enhancement, digital cameras, medical imaging

For an image that is too bright, Gamma (Power-Law) Transformation with $\gamma > 1$ is the most suitable grey level transformation.

2) A Dark Image Lack Details

The most suitable grey level transformation for a dark image is logarithmic (log) Transformation. It enhances the visibility of the low intensity (dark) regions by expandings their grey level value while compressing higher intensity value

Grey level Transformation : Log Transformation

$$S = c\log(1+r)$$

Application process :-

1. Read the input image
2. Normalize pixel value if required
3. Apply the logarithmic transformation
   $S = c\log(1+r)$
4. Scale the transformed value back to the range 0-255
5. Display the enhanced image

Justification:-

→ It greatly enhances low intensity pixels
→ It preserves bright regions by compressing high intensity values
→ It is widely used in medical imaging, satellite image processing and low-light photography to reveal hidden detail

3. An image has Poor constrast

The most suitable historgram-based technique for improving the contrast of an image is histogram Equalization. It redistributes the intensity value so that the histogram is spread over the entire grey level range level range, resulting in better contrast and improved visibility.

Histogram Equalization

A histogram represent the distribution of pixel intensities in a image. If the histogram is concentrated within a small range of gray level, the image has poor contrast. Histogram Equalization redistributes these pixel intensities across the full range (0-255), making dark area darker and bright area brighter where needed

Outcome:-

contrast is significantly improved
Fine detail in both dark and bright regions become more visible.

For an image with poor contrast, Histogram Equalization is the most appropriate histogram-based enhancement technique. It redistributes pixel intensities across the full grey level range, resulting in higher contrast, improved visibility

4) A Low Contrast Image Requires Uniform Intensity Distribution:-

Histogram Equalization is a histogram-based image enhancement technique that increase the global constrast of an image. It spread the pixel intensites over the full available range (0-255)

Working process:-

1. Compute the histogram of the input image
2. Calculate the CDF from the histogram
3. Normalize the CDF to map pixels
4. Replace each original pixel intensity
5. Obtain the enhanced image with a more uniform intensity

Outcome:-

Improved overall image contrast
Better visibility of object and details
Enhanced image quality for visual analysis
Uniform utilization

5) An image show uneven brightness across region

The most appropriate enhancement technique is Adaptive Histogram Equalization or its improved version Contrast Limited Adaptive Histogram Equalization (CLAHE). These techniques enhance constrast locally, making them ideal for ~~cottae~~ correcting non-uniform illumination.

Concept :-

Unlike standard Histogram Equalization, which process the entire image at once, CLAHE divides the image into small region (titles) and applies histogram equalization to each region separately. It also limits the constrast to prevent excessive noise amplification

Working process :-

1. Dividing the image into small, non-overlapping tiles.
2. Compute the histogram for each tile.
3. Apply histogram equalization
4. Limit the constrast using a predefined clip limit
5. Interpolate between more visible throughout the image

Outcome :-

- Uniform illunation across the image
- Enhanced local contrast
- Improved visibility of edges and textures

Justification

CLAHE is preffered because

- It effectively correct non-uniform illumination
- It enhancement local details without producing excessive brightness.
- It minimizes noise amplification compared to Standard Adaptive Histogram Equalization.

For an image with uneven brightness across regions Contrast Limited Adaptive Histogram Equalization is the most suitable enhancement technique. It improves local contrast, correct non-uniform illumination, preserves image detail and produces a balanced high quality image